INCH-POUND

MIL-C-24640/7C
26 June 1995
SUPERSEDING
MIL-C-24640/7B
31 July 1989

MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT, MULTI-CONDUCTOR, AWG 22, TYPE 1XMSO

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION
(Non-Watertight)

FIRST - 19 strand copper conductor in accordance with ASTM B-286, tinned strands in accordance with ASTM B-33. AWG size 22 Maximum diameter 0.033 inches.

SECOND - Component wires shall be MIL-W-81044/12 and shall be tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with Table VI of MIL-C-24640.

THIRD - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90% determined in accordance with MIL-C-27500.

FOURTH - Shield insulation of two polyester tapes or a polyester tape and extruded jacket. Identification codes applied using Method 2.

FIFTH - The required number of shielded and jacketed singles shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

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CONSTRUCTION
(Non-Watertight) continued:

SIXTH - Binder tape applied helically with overlap.

SEVENTH - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.

EIGHTH - Binder tape applied helically with overlap, at manufacturer's option.

NINTH - Crosslinked polyolefin jacket. (See Table I for thickness).

TABLE I. Construction details.

PIN M24640/7	Type number	Number of component wires	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01UO	1XMSO-7	7	0.247	0.038	0.344	0.370	116.0
-02UO	1XMSO-16	16	.379	.041	.471	.507	212.0
-03UO	1XSMO-70	70	.785	.044	.859	.925	735.0

REQUIREMENTS:

Qualification required.
Construction details: See table I.

Component wire:

For the component wire the following additional tests shall be applicable in accordance with MIL-W-81044:

WIRE, RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300± 2°C (572± 3.6°F) for 6 hours.
Blocking: 150± 2°C (302± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test): 6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak) or 3 Khz spark test 5.5 KV, 100 percent test.

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Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200± 2°C (392± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300± 2°C (572± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

	Basic electrical:	Requirements
	Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.11
	Voltage withstand - volts, root mean square, minimum	
	Conductor to conductor	Not applicable
	Conductor to shield	1000
	Shield to shield	200
	Component shield to overall shield	500
*	Insulation resistance - megohms/1000 feet, minimum	
	Conductor to shield	2500
	Shield to shield (component shield only)	100
	Shield to shield (component shield to overall shield)	100
	Conductor and shield continuity	No failure
	Jacket flaws	No failure
	Group A:	
	Visual and dimensional	No failure
	Capacitance:	
	Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	120
	Capacitance unbalance - percent, maximum	Not applicable
	Characteristic impedance - at 1 MHz, ohms	13 min, 30 max
	Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
	Weight	No failure
	Group B:	
	Cold bending	No failure
*	Cross link proof test (percent maximum)	
	Jacket	50
*	Drip	Not applicable
	Tear - pounds per inch thickness, minimum	35

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Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	4
Gas flame (1 hour)	Not applicable

Group C:

Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged minimum	50
* Elongation - percent of unaged, minimum	50
* Permanence of printing (conductor) - cycles, minimum	Not applicable
* Permanence of printing (jacket) - cycles, minimum	125
Armor - conformance to material, construction and coverage	No failure
* Shield - conformance to material, construction and coverage	Not applicable
Heat distortion - percent of unaged, maximum	30
* Cable sealant removability	Not applicable
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
* Surface transfer impedance	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

* Group D:	
Flame propagation	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility cable	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50

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*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10
*	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
	Gas Flame (3 hour)	Not applicable
	Component wire	No failure

Unit ordering length: 7 & 16 500 feet nominal.
70 250 feet nominal.

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